# Table of Contents

- Governing Equations
    - Momentum Equation
    - Continuity Equation
    - Turbulence Equation
- Reconstruction Scheme
    - Face Reconstruction
        - Linear Interpolation
        - Non-Orthogonal Correction
        - Skewness Correction
    - Gradient Reconstruction
        - Green-Gauss Gradient
        - Least Squares Gradient
    - Reconstruction Algorithm
- Boundary Conditions
- Transport Equation
    - Diffusion Term
        - Minimum Correction
        - Orthogonal Correction
        - Over-Relaxed
        - Matrix Assembly
    - Convection Term
        - Barth–Jespersen Limiter
        - Venkatakrishnan Limiter
        - Matrix Assembly
    - Temporal Term
        - Matrix Assembly
    - Source Term
        - Matrix Assembly
- Pressure Equation
    - Matrix Assembly
- Turbulence Model
    - Spalart-Allmaras Model
    - Wall Function
        - Spalding's Law
        - Newton-Raphson Iteration
    - Source Assembly
- Solver Algorithm
- Appendix

## Governing Equations

### Momentum Equation

$$\underbrace{\frac{\partial \phi}{\partial t}}_{\text{temporal}} + \underbrace{\nabla \cdot (\phi \mathbf{u})}_{\text{convection}} = \underbrace{\nabla \cdot [(\nu + \nu_t) \nabla \phi]}_{\text{diffusion}} - \underbrace{\nabla p \cdot \hat{\mathbf{e}}_\phi}_{\text{source}}$$

$$\phi \in \{u, v\} \qquad \nu = \frac{1}{\mathrm{Re}} \qquad \nu_{\text{eff}} = \nu + \nu_t \qquad \hat{\mathbf{e}}_\phi \in \{\hat{\mathbf{i}}, \hat{\mathbf{j}}\}$$

$\nu_t$ is supplied by the Spalart-Allmaras model; $\nu_t = 0$ recovers the laminar solver.

### Continuity Equation

$$\nabla \cdot \mathbf{u} = 0$$

### Turbulence Equation

$$\underbrace{\frac{\partial \tilde{\nu}}{\partial t}}_{\text{temporal}} + \underbrace{\nabla \cdot (\tilde{\nu}\mathbf{u})}_{\text{convection}} = \underbrace{\nabla \cdot (\frac{\nu + \tilde{\nu}}{\sigma} \nabla \tilde{\nu})}_{\text{diffusion}} + \underbrace{c_{b1} \tilde{S} \tilde{\nu}}_{\text{production}} - \underbrace{c_{w1} f_w \left(\frac{\tilde{\nu}}{d}\right)^2}_{\text{destruction}} + \underbrace{\frac{c_{b2}}{\sigma} \|\nabla \tilde{\nu}\|^2}_{\text{cross-diffusion}}$$

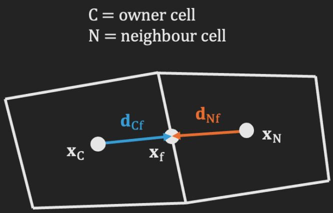
$$\tilde{\nu} \leftarrow \max(\tilde{\nu}, 0)$$

$\tilde{S}$, $f_w$, and $d$ are defined in Spalart-Allmaras model.

## Reconstruction Scheme

### Face Reconstruction

#### Linear Interpolation


C = owner cell
N = neighbour cell
$\mathbf{d}_{\mathrm{Cf}}$
$\mathbf{d}_{\mathrm{Nf}}$
$\mathbf{x}_{\mathrm{C}}$
$\mathbf{x}_{\mathrm{f}}$
$\mathbf{x}_{\mathrm{N}}$

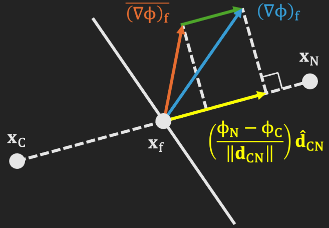
$$\overline{\phi_f} = \begin{cases} w_f \phi_C + (1 - w_f) \phi_N & \text{interior face} \\ \phi_C & \text{boundary face} \end{cases}$$

$$\overline{(\nabla \phi)_f} = \begin{cases} w_f (\nabla \phi)_C + (1 - w_f)(\nabla \phi)_N & \text{interior face} \\ (\nabla \phi)_C & \text{boundary face} \end{cases}$$

$$w_f = \frac{\|\mathbf{d}_{Nf}\|}{\|\mathbf{d}_{Cf}\| + \|\mathbf{d}_{Nf}\|}$$

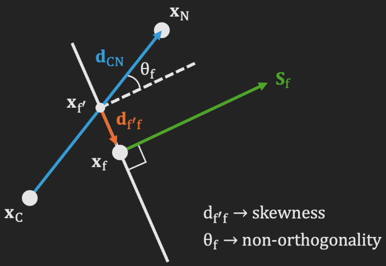
#### Non-Orthogonal Correction

$$\left(\frac{\phi_{\mathrm{N}} - \phi_{\mathrm{C}}}{\|\mathbf{d}_{\mathrm{CN}}\|} - \overline{(\nabla \phi)_{\mathrm{f}}} \cdot \hat{\mathbf{d}}_{\mathrm{CN}}\right) \hat{\mathbf{d}}_{\mathrm{CN}}$$


$\overline{(\nabla\phi)}_f$
$(\nabla\phi)_f$
$\mathbf{x}_N$
$\mathbf{x}_C$
$\mathbf{x}_f$
$\left(\frac{\phi_N-\phi_C}{\|\mathbf{d}_{CN}\|}\right)\hat{\mathbf{d}}_{CN}$

$$(\nabla\phi)_f=\overline{(\nabla\phi)}_f+\left[\frac{\phi_N-\phi_C}{\|\mathbf{d}_{CN}\|}-\overline{(\nabla\phi)}_f\cdot\hat{\mathbf{d}}_{CN}\right]\hat{\mathbf{d}}_{CN}$$

### Skewness Correction



$$\phi_f=\overline{\phi_f}+(\nabla\phi)_f\cdot\mathbf{d}_{f'f}$$

## Gradient Reconstruction

### Green-Gauss Gradient

$$(\nabla\phi)_C=\frac{1}{V_C}\sum_f\phi_f\mathbf{S}_f$$

### Least Squares Gradient

$$J=\sum_N w_{CN}^2\left[(\nabla\phi)_C\cdot\mathbf{d}_{CN}-(\phi_N-\phi_C)\right]^2$$

$$w_{CN}=\frac{1}{\|\mathbf{d}_{CN}\|}$$

$$\mathbf{M}_C(\nabla\phi)_C=\mathbf{r}_C$$

$$\mathbf{M}_C=\sum_N w_{CN}^2\mathbf{d}_{CN}\mathbf{d}_{CN}^\top\qquad\mathbf{r}_C=\sum_N w_{CN}^2(\phi_N-\phi_C)\mathbf{d}_{CN}$$

$$(\nabla\phi)_C=\mathbf{M}_C^{-1}\mathbf{r}_C$$

## Reconstruction Algorithm

1. Interpolate face scalar $\overline{\phi}_f$.
2. Apply boundary conditions.

3. Compute cell gradient $\overline{(\nabla\phi)_C}$.
4. Interpolate face gradient $\overline{(\nabla\phi)_f}$.
5. Correct non-orthogonality for face gradient $(\nabla\phi)_f$.
6. Correct skewness for face scalar $\phi_f$.
7. Apply boundary conditions.

## Boundary Conditions

| Field | Body Face | Farfield Face (Inflow, $\dot{m}_f \leq 0$) | Farfield Face (Outflow, $\dot{m}_f > 0$) |
|---|---|---|---|
| $\mathbf{u}$ | $\mathbf{0}$ | $\mathbf{u}_\infty$ | extrapolated from cell |
| $\mathbf{h}$ | $\mathbf{0}$ | $\mathbf{u}_\infty$ | extrapolated from cell |
| $p$ | extrapolated from cell | extrapolated from cell | $0$ |
| $\dot{m}_f$ | $0$ | $\mathbf{u}_\infty \cdot \mathbf{S}_f$ | from pressure correction |
| $\tilde{\nu}$ | $0$ | $\tilde{\nu}_\infty$ | extrapolated from cell |
| $\nu + \nu_t$ | $\nu$ (integrate to wall)<br>$\nu_\text{wall}$ (wall function) | extrapolated from cell | extrapolated from cell |

## Transport Equation

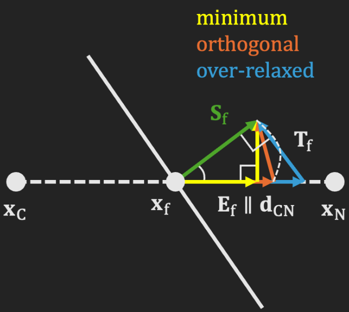
The momentum and turbulence equations share the form

$$\underbrace{\frac{\partial\phi}{\partial t}}_{\text{temporal}} + \underbrace{\nabla\cdot(\phi\mathbf{u})}_{\text{convection}} - \underbrace{\nabla\cdot(\Gamma\nabla\phi)}_{\text{diffusion}} = \underbrace{S_\phi}_{\text{source}}$$

with the following specialisations:

| Equation | $\phi$ | $\Gamma$ | $S_\phi$ |
|---|---|---|---|
| Momentum | $u, v$ | $\nu + \nu_t$ | $-\nabla p \cdot \hat{\mathbf{e}}_\phi$ |
| Turbulence | $\tilde{\nu}$ | $(\nu + \tilde{\nu})/\sigma$ | $c_{b1}\tilde{S}\tilde{\nu} - c_{w1} f_w (\tilde{\nu}/d)^2 + (c_{b2}/\sigma)\lVert\nabla\tilde{\nu}\rVert^2$ |

### Diffusion Term



$$\int_V \nabla\cdot(\Gamma\nabla\phi)dV = \sum_f \Gamma_f(\nabla\phi)_f\cdot\mathbf{S}_f$$

$$\mathbf{S}_f = \mathbf{E}_f + \mathbf{T}_f$$

$$\Gamma_f(\nabla\phi)_f\cdot\mathbf{S}_f = \underbrace{\Gamma_f||\mathbf{E}_f||\frac{\phi_N-\phi_C}{||\mathbf{d}_{CN}||}}_{\text{implicit}} + \underbrace{\Gamma_f(\nabla\phi)_f\cdot\mathbf{T}_f}_{\text{explicit}}$$

### Minimum Correction

$$\mathbf{E}_f = (\mathbf{S}_f\cdot\hat{\mathbf{d}}_{CN})\hat{\mathbf{d}}_{CN}$$

### Orthogonal Correction

$$\mathbf{E}_f = ||\mathbf{S}_f||\hat{\mathbf{d}}_{CN}$$

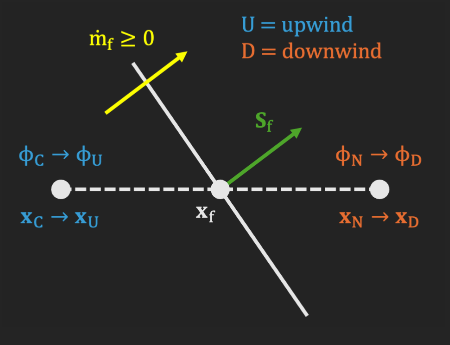
### Over-Relaxed

$$\mathbf{E}_f = \frac{||\mathbf{S}_f||^2}{\mathbf{S}_f\cdot\hat{\mathbf{d}}_{CN}}\hat{\mathbf{d}}_{CN}$$

### Matrix Assembly

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|
| Interior Face | $\mathbf{A}[C,C], \mathbf{A}[N,N] \mathrel{+}= +\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{CN}\rVert$<br>$\mathbf{A}[C,N], \mathbf{A}[N,C] \mathrel{+}= -\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{CN}\rVert$ | $\mathbf{b}[C] \mathrel{+}= +\Gamma_f(\nabla\phi)_f\cdot\mathbf{T}_f$<br>$\mathbf{b}[N] \mathrel{+}= -\Gamma_f(\nabla\phi)_f\cdot\mathbf{T}_f$ |
| Body Face | $\mathbf{A}[C,C] \mathrel{+}= +\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{Cf}\rVert$ | $\mathbf{b}[C] \mathrel{+}= +\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{Cf}\rVert\cdot\phi_f + \Gamma_f(\nabla\phi)_f\cdot\mathbf{T}_f$ |
| Farfield Face (Inflow) | $\mathbf{A}[C,C] \mathrel{+}= +\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{Cf}\rVert$ | $\mathbf{b}[C] \mathrel{+}= +\Gamma_f\lVert\mathbf{E}_f\rVert/\lVert\mathbf{d}_{Cf}\rVert\cdot\phi_f + \Gamma_f(\nabla\phi)_f\cdot\mathbf{T}_f$ |
| Farfield Face (Outflow) | $-$ | $-$ |

## Convection Term



$$\int_V \nabla\cdot(\phi\mathbf{u})dV = \sum_f \dot{m}_f\phi_f$$

$$\dot{m}_f = \mathbf{u}_f\cdot\mathbf{S}_f$$

$$U = \begin{cases} C & \dot{m}_f \geq 0 \\ N & \dot{m}_f < 0 \end{cases}$$

$$\dot{m}_f \phi_f = \underbrace{\dot{m}_f \phi_U}_{\text{implicit}} + \underbrace{\dot{m}_f \Psi_U (\nabla \phi)_U \cdot \mathbf{d}_{Uf}}_{\text{explicit}}$$

$$\Psi_U \in [0,1] \qquad \Delta_f = (\nabla \phi)_U \cdot \mathbf{d}_{Uf}$$

$$\phi_{\max} = \max(\phi_U, \max_N \phi_N) \qquad \phi_{\min} = \min(\phi_U, \min_N \phi_N)$$

$$\Delta_{\max} = \phi_{\max} - \phi_U \qquad \Delta_{\min} = \phi_{\min} - \phi_U$$

**Barth–Jespersen Limiter**

$$\Psi_U = \min_f \begin{cases} \min\left(1, \dfrac{\Delta_{\max}}{\Delta_f}\right) & \Delta_f > 0 \\ \min\left(1, \dfrac{\Delta_{\min}}{\Delta_f}\right) & \Delta_f < 0 \\ 1 & \Delta_f = 0 \end{cases}$$

**Venkatakrishnan Limiter**

$$\Delta_* = \begin{cases} \Delta_{\max} & \Delta_f > 0 \\ \Delta_{\min} & \Delta_f < 0 \end{cases}$$

$$K \in [1,5]$$

$$\epsilon^2 = (K V_U^{1/n_{\text{dim}}})^3$$

$$\Psi_U = \min_f \frac{\Delta_*^2 + 2\Delta_f \Delta_* + \epsilon^2}{\Delta_*^2 + 2\Delta_f^2 + \Delta_f \Delta_* + \epsilon^2}$$

**Matrix Assembly**

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|
| Interior Face | $\mathbf{A}[C,C] \mathrel{+}= +\max(\dot{m}_f, 0)$<br>$\mathbf{A}[C,N] \mathrel{+}= +\min(\dot{m}_f, 0)$<br>$\mathbf{A}[N,N] \mathrel{+}= -\min(\dot{m}_f, 0)$<br>$\mathbf{A}[N,C] \mathrel{+}= -\max(\dot{m}_f, 0)$ | $\mathbf{b}[C] \mathrel{+}= -\dot{m}_f \Psi_U (\nabla \phi)_U \cdot \mathbf{d}_{Uf}$<br>$\mathbf{b}[N] \mathrel{+}= +\dot{m}_f \Psi_U (\nabla \phi)_U \cdot \mathbf{d}_{Uf}$ |
| Body Face | – | – |
| Farfield Face (Inflow) | – | $\mathbf{b}[C] \mathrel{+}= -\min(\dot{m}_f, 0) \cdot \phi_f$ |
| Farfield Face (Outflow) | $\mathbf{A}[C,C] \mathrel{+}= +\max(\dot{m}_f, 0)$ | $\mathbf{b}[C] \mathrel{+}= -\max(\dot{m}_f, 0) \Psi_C (\nabla \phi)_C \cdot \mathbf{d}_{Cf}$ |

## Temporal Term

$$\int_V \frac{\partial \phi}{\partial t} dV \approx V_C \frac{\phi_C^{(n+1)} - \phi_C^{(n)}}{\Delta t}$$

**Matrix Assembly**

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|
| Cell | $\mathbf{A}[C,C] \mathrel{+}= V_C / \Delta t$ | $\mathbf{b}[C] \mathrel{+}= (V_C / \Delta t) \phi_C^{(n)}$ |

## Source Term

$$\int_V S_\phi dV \approx V_C S_\phi$$

**Matrix Assembly**

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|

| | | |
|---|---|---|
| Cell | – | $\mathbf{b}[C] \mathrel{+}= V_C S_\phi$ |

## Pressure Equation

* see Appendix for more information

$$\nabla \cdot (\Gamma \nabla p) = \nabla \cdot \mathbf{h}$$

$$\Gamma_C = \frac{V_C}{a_C} \qquad \mathbf{h}_C = \mathbf{u}_C + \frac{(\mathbf{b}_{-p})_C - (\mathbf{A}\mathbf{u})_C}{a_C}$$

$$\int_V \nabla \cdot (\Gamma \nabla p) dV = \sum_f \Gamma_f (\nabla p)_f \cdot \mathbf{S}_f$$

$$\int_V \nabla \cdot \mathbf{h} dV = \sum_f \mathbf{h}_f \cdot \mathbf{S}_f$$

$$\dot{m}_f^* = \begin{cases} \mathbf{h}_f \cdot \mathbf{S}_f + \frac{\Gamma_f}{\Delta t}\left(\dot{m}_f^{(n)} - \mathbf{u}_f^{(n)} \cdot \mathbf{S}_f\right) & \text{interior face} \\ \mathbf{h}_f \cdot \mathbf{S}_f & \text{boundary face} \end{cases}$$

### Matrix Assembly

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|
| Interior Face | $\mathbf{A}[C,C], \mathbf{A}[N,N] \mathrel{+}= +\Gamma_f \Vert\mathbf{E}_f\Vert / \Vert\mathbf{d}_{CN}\Vert$<br>$\mathbf{A}[C,N], \mathbf{A}[N,C] \mathrel{+}= -\Gamma_f \Vert\mathbf{E}_f\Vert / \Vert\mathbf{d}_{CN}\Vert$ | $\mathbf{b}[C] \mathrel{+}= -\dot{m}_f^* + \Gamma_f (\nabla p)_f \cdot \mathbf{T}_f$<br>$\mathbf{b}[N] \mathrel{+}= +\dot{m}_f^* - \Gamma_f (\nabla p)_f \cdot \mathbf{T}_f$ |
| Body Face | – | – |
| Farfield Face (Inflow) | – | $\mathbf{b}[C] \mathrel{+}= -\mathbf{u}_\infty \cdot \mathbf{S}_f$ |
| Farfield Face (Outflow) | $\mathbf{A}[C,C] \mathrel{+}= +\Gamma_f \Vert\mathbf{E}_f\Vert / \Vert\mathbf{d}_{Cf}\Vert$ | $\mathbf{b}[C] \mathrel{+}= -\mathbf{h}_f \cdot \mathbf{S}_f + \Gamma_f (\nabla p)_f \cdot \mathbf{T}_f$ |

## Turbulence Model

### Spalart-Allmaras Model

$$\chi = \frac{\tilde{\nu}}{\nu} \qquad f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3} \qquad f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}} \qquad \nu_t = \tilde{\nu} f_{v1}$$

$$\Omega = \left| \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right| \qquad \tilde{S} = \max\left(\Omega + \frac{\tilde{\nu} f_{v2}}{(\kappa d)^2},\ c_s \Omega\right) \qquad d \equiv \text{nearest wall distance}$$

$$r = \min\left(\frac{\tilde{\nu}}{\tilde{S}(\kappa d)^2},\ r_{\max}\right) \qquad g = r + c_{w2}(r^6 - r) \qquad f_w = g\left(\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right)^{1/6}$$

| Constant | Value |
|---|---|
| $\sigma$ | $2/3$ |
| $\kappa$ | $0.41$ |
| $c_{b1}$ | $0.1355$ |
| $c_{b2}$ | $0.622$ |
| $c_{v1}$ | $7.1$ |
| $c_{w1}$ | $c_{b1}/\kappa^2 + (1 + c_{b2})/\sigma$ |
| $c_{w2}$ | $0.3$ |

| | |
|---|---|
| $c_{w3}$ | $2.0$ |
| $c_s$ | $0.3$ |
| $r_{\max}$ | $10$ |

### Wall Function

#### Spalding's Law

(implicit in $u_\tau$)

$$y^+ = u^+ + \frac{1}{E}\left[\exp(\kappa u^+) - 1 - \kappa u^+ - \frac{(\kappa u^+)^2}{2} - \frac{(\kappa u^+)^3}{6}\right]$$

$$y^+ = \frac{u_\tau y_c}{\nu} \qquad u^+ = \frac{u_\parallel}{u_\tau} \qquad E = 9.8$$

#### Newton-Raphson Iteration

(solve for $u_\tau$)

$$u_\tau^{(0)} = \sqrt{\frac{\nu u_\parallel}{y_c}} \qquad u_\tau \leftarrow \max(u_\tau - f/f',\ \epsilon)$$

$$f(u_\tau) = y^+ - u^+ - \frac{1}{E}\left[\exp(\kappa u^+) - 1 - \kappa u^+ - \frac{(\kappa u^+)^2}{2} - \frac{(\kappa u^+)^3}{6}\right]$$

$$f'(u_\tau) = \frac{y_c}{\nu} + \frac{u_\parallel}{u_\tau^2} + \frac{\kappa u^+\left[\exp(\kappa u^+) - 1 - \kappa u^+ - (\kappa u^+)^2/2\right]}{E\,u_\tau}$$

$$\nu_{\text{wall}} = \frac{u_\tau^2 y_c}{u_\parallel}$$

### Source Assembly

Temporal, convection, and diffusion follow the Transport Equation assembly with $\phi = \tilde{\nu}$ and $\Gamma_f = (\nu + \tilde{\nu}_f)/\sigma$. Production and cross-diffusion are added to $\mathbf{b}$, while destruction is linearised semi-implicitly to put a positive coefficient on the $\mathbf{A}$ diagonal.

| Element | $\mathbf{A}$ Contribution | $\mathbf{b}$ Contribution |
|---|---|---|
| Cell (Production) | $-$ | $\mathbf{b}[C] \mathrel{+}= +c_{b1}\tilde{S}_C\tilde{\nu}_C V_C$ |
| Cell (Destruction) | $\mathbf{A}[C,C] \mathrel{+}= +c_{w1} f_{w,C}\,\tilde{\nu}_C V_C/d_C^2$ | $-$ |
| Cell (Cross-Diffusion) | $-$ | $\mathbf{b}[C] \mathrel{+}= +(c_{b2}/\sigma)\|(\nabla\tilde{\nu})_C\|^2 V_C$ |

## Solver Algorithm

Initialise $\mathbf{u}, p, \dot{m}_f$, and (if turbulence is enabled) $\tilde{\nu}, \nu_t$ from the initial and boundary conditions.

For each time step $n$:

1. Save $\dot{m}_f^{(n)} \leftarrow \dot{m}_f$; reclassify each farfield face as inflow ($\dot{m}_f \le 0$) or outflow ($\dot{m}_f > 0$).
2. **Momentum predictor**:
    1. For $\phi \in \{u, v\}$: reconstruct $\phi_f$ from $\phi$; save $\phi_f^{(n)} \leftarrow \phi_f$.
    2. Assemble $\mathbf{A}$ (shared by $u, v$) from temporal, diffusion ($\Gamma_f = \nu + \nu_t$), and convection.
    3. For $\phi \in \{u, v\}$:
        1. Assemble $\mathbf{b}$ from temporal, diffusion, and convection; save $\mathbf{b}_{-p} \leftarrow \mathbf{b}$.
        2. Add source $-\nabla p \cdot \hat{\mathbf{e}}_\phi$ to $\mathbf{b}$; solve $\mathbf{A}\phi = \mathbf{b}$.
3. Compute $\Gamma_C = V_C/\mathbf{A}_{CC}$; reconstruct $\Gamma_f$ from $\Gamma$.

4. Assemble pressure matrix $\mathbf{A}_p$ from $\Gamma_f$.

5. **Pressure corrector** (repeat $n_{\text{piso}}$ times):

    1. Compute $\mathbf{h}_C = \mathbf{u}_C + ((\mathbf{b}_{-p})_C - (\mathbf{A}\mathbf{u})_C)/\mathbf{A}_{CC}$; reconstruct $\mathbf{h}_f$ from $\mathbf{h}$.
    2. Compute $\dot{m}_f^*$ from $\mathbf{h}_f$.
    3. Assemble $\mathbf{b}$; solve $\mathbf{A}_p p = \mathbf{b}$.
    4. Correct $\dot{m}_f \leftarrow \dot{m}_f^* - \Gamma_f (\nabla p)_f \cdot \mathbf{S}_f$; reconstruct $p_f$ from $p$.
    5. Correct $\mathbf{u}_C \leftarrow \mathbf{h}_C - \Gamma_C (\nabla p)_C$.

6. **Turbulence corrector** (if enabled):

    1. Reconstruct $\tilde{\nu}_f$ from $\tilde{\nu}$; evaluate closure $\tilde{S}, f_w$ from $\tilde{\nu}^{(n)}, \Omega, d$.
    2. Assemble $\mathbf{A}_{\tilde{\nu}}$ from temporal, diffusion ($\Gamma_f = (\nu + \tilde{\nu}_f)/\sigma$), convection, and destruction.
    3. Assemble $\mathbf{b}$ from temporal, diffusion, convection, production, and cross-diffusion.
    4. Solve $\mathbf{A}_{\tilde{\nu}} \tilde{\nu} = \mathbf{b}$; clamp $\tilde{\nu} \leftarrow \max(\tilde{\nu}, 0)$.
    5. Update $\nu_t = \tilde{\nu} f_{v1}$; refresh $(\nu_{\text{eff}})_f = \nu + (\nu_t)_f$ with body-face override ($\nu$ or $\nu_{\text{wall}}$).

## Appendix

### Pressure Equation

Let $\mathbf{u}_C := [u_C, v_C]^\top$ denote the cell-centred velocity. The discrete momentum equation at cell $C$ separates the pressure source from the rest of the right-hand side:

$$a_C \mathbf{u}_C + \sum_{N \in C} a_N \mathbf{u}_N = \mathbf{b}_C = (\mathbf{b}_{-p})_C - (\nabla p)_C V_C$$

Solving for $\mathbf{u}_C$ and grouping the pressure-free part as $\mathbf{h}_C$ with prefactor $\Gamma_C := V_C / a_C$:

$$\mathbf{u}_C = \frac{(\mathbf{b}_{-p})_C}{a_C} - \sum_{N \in C} \frac{a_N}{a_C} \mathbf{u}_N - \frac{V_C}{a_C} (\nabla p)_C$$

$$\mathbf{h}_C := \frac{(\mathbf{b}_{-p})_C}{a_C} - \sum_{N \in C} \frac{a_N}{a_C} \mathbf{u}_N \qquad \Gamma_C := \frac{V_C}{a_C}$$

$$\mathbf{u}_C = \mathbf{h}_C - \Gamma_C (\nabla p)_C$$

Note that the off-diagonal sum in $\mathbf{h}_C$ can be rewritten using the matrix-vector product $\mathbf{A}\mathbf{u}$. Since

$$(\mathbf{A}\mathbf{u})_C = a_C \mathbf{u}_C + \sum_{N \in C} a_N \mathbf{u}_N$$

we have

$$\sum_{N \in C} \frac{a_N}{a_C} \mathbf{u}_N = \frac{(\mathbf{A}\mathbf{u})_C}{a_C} - \mathbf{u}_C$$

Substituting into the definition of $\mathbf{h}_C$:

$$\mathbf{h}_C = \frac{(\mathbf{b}_{-p})_C}{a_C} - \sum_{N \in C} \frac{a_N}{a_C} \mathbf{u}_N = \mathbf{u}_C + \frac{(\mathbf{b}_{-p})_C - (\mathbf{A}\mathbf{u})_C}{a_C}$$

The code uses this latter form because $\mathbf{A}\mathbf{u}$ is a single matrix-vector multiplication.

Vectorising over all cells and imposing continuity $\nabla \cdot \mathbf{u} = 0$ yields the pressure Poisson equation:

$$\mathbf{u} = \mathbf{h} - \Gamma \nabla p$$

$$\therefore \nabla \cdot (\Gamma \nabla p) = \nabla \cdot \mathbf{h}$$

Solving this Poisson equation gives $p$, which projects $\mathbf{u}$ onto the divergence-free constraint via $\mathbf{u} = \mathbf{h} - \Gamma \nabla p$.

### Rhie-Chow Residual

On a **staggered** grid the face mass flux is updated directly via the face-centred relation:

$$\dot{m}_f \leftarrow \dot{m}_f^* - \Gamma_f (\nabla p)_f \cdot \mathbf{S}_f$$

On a **collocated** grid the cell velocity is updated and then interpolated to the face:

$$\mathbf{u}_C \leftarrow \mathbf{h}_C - \Gamma_C (\nabla p)_C$$

$$\mathbf{u}_f \leftarrow w_f \mathbf{u}_C + (1 - w_f) \mathbf{u}_N$$

The face flux from this interpolation differs from the face-centred update:

$$\dot{m}_f - \mathbf{u}_f \cdot \mathbf{S}_f \neq 0$$

Rhie-Chow removes this residual by computing $\dot{m}_f$ via a face-centred update directly, recovering staggered-grid behaviour on a collocated mesh.

## Rhie-Chow Correction

For unsteady flow, the predictor RHS without the pressure source contains a temporal contribution:

$$(\mathbf{b}_{-p})_C = \frac{V_C}{\Delta t} \mathbf{u}_C^{(n)} + \text{convection} + \text{diffusion}$$

Dividing by $a_C$ produces a corresponding $\Gamma_C / \Delta t$ term in $\mathbf{h}_C$:

$$\mathbf{h}_C = \frac{(\mathbf{b}_{-p})_C}{a_C} - \sum_{N \in C} \frac{a_N}{a_C} \mathbf{u}_N = \frac{\Gamma_C}{\Delta t} \mathbf{u}_C^{(n)} + \text{rest}$$

Interpolating to the face and projecting onto $\mathbf{S}_f$:

$$\mathbf{h}_f = \frac{\Gamma_f}{\Delta t} \mathbf{u}_f^{(n)} + (\text{rest})_f$$

$$\mathbf{h}_f \cdot \mathbf{S}_f = \frac{\Gamma_f}{\Delta t} \left( \mathbf{u}_f^{(n)} \cdot \mathbf{S}_f \right) + (\text{rest})_f \cdot \mathbf{S}_f$$

Replacing the collocated face-velocity contribution $\mathbf{u}_f^{(n)} \cdot \mathbf{S}_f$ with the face-centred mass flux $\dot{m}_f^{(n)}$ from the previous step gives the Rhie-Chow-corrected predictor flux:

$$\therefore \dot{m}_f^* \leftarrow \mathbf{h}_f \cdot \mathbf{S}_f + \frac{\Gamma_f}{\Delta t} \left( \dot{m}_f^{(n)} - \mathbf{u}_f^{(n)} \cdot \mathbf{S}_f \right)$$

The correction $\dot{m}_f^{(n)} - \mathbf{u}_f^{(n)} \cdot \mathbf{S}_f$ replaces the collocated-interpolated face velocity contribution with the face-centred (Rhie-Chow-corrected) mass flux from the previous step.